

Log Analysis: Automated Error Clustering & Signature Libraries

Duration : 90 Min.

Scenario

You need to write a Python script that reads raw application logs, normalizes stack trace signatures, clusters identical errors, and matches them to runbook entries.

Task Objectives

Perform the following actions inside the `student_workspace` directory: - Complete the Python script 'cluster_errors.py' to produce 'error_clusters.json'.

Instructions to Perform the Task

1. When your workspace loads in **VS Code**, use the **Explorer** panel on the left to locate your files.
2. Navigate to and click the specific files mentioned in the Task Objectives.
3. Make the necessary code edits in the editor.
4. Press `Ctrl + S` (Windows) or `Cmd + S` (Mac) to save your changes.
5. If you need to run commands, open the built-in terminal by clicking **Terminal > New Terminal** from the top menu.
6. Verify your progress by running `python run.py` locally in the terminal.

Validation

Once you have saved your files and verified your progress, return to the platform dashboard and click the **"Run Test" / "Verify"** button. This will automatically evaluate your changes and generate your score!

Grading Criteria

Your performance will be evaluated based on the following test cases:

Test Case	Requirement	Marks
TC1	cluster_errors.py is syntax-valid Python	3 Marks
TC2	Parses error logs and normalizes dynamic IDs/timestamps	3 Marks
TC3	Groups identical error signatures into clusters	3 Marks

TC4	Writes structured JSON output to error_clusters.json	3 Marks
TC5	Matches cluster signature to runbook ID	2 Marks
TC6	Calculates error frequency per cluster	2 Marks
TC7	Handles missing/malformed log lines gracefully	2 Marks
TC8	Includes first and last occurrence timestamps	2 Marks

Total Score: 20 Marks

Important Notes

- This is an auto-evaluated question. Ensure all code edits are properly saved and the 'run.py' checks pass before submission.